

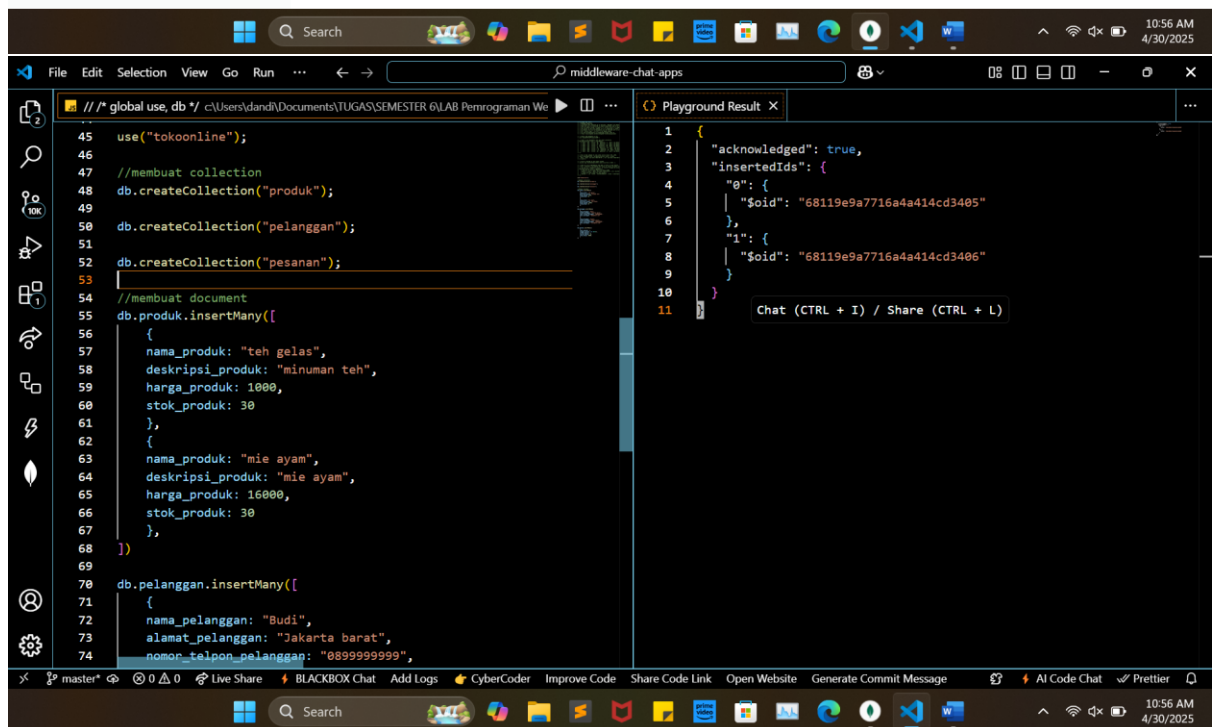
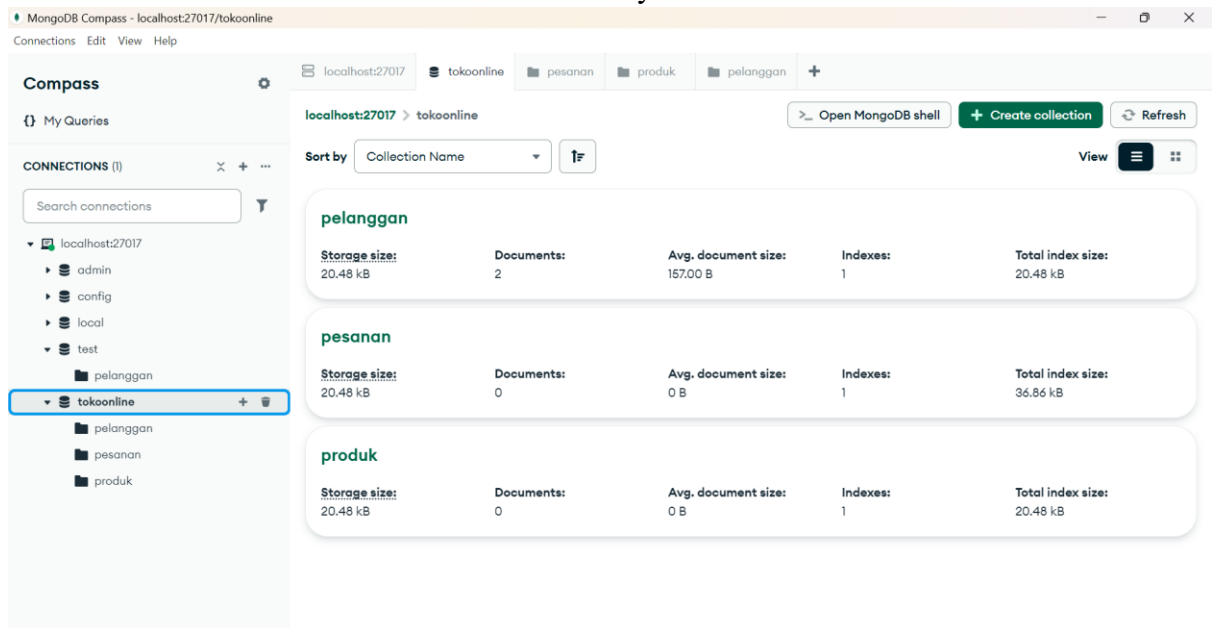
Nama Kelompok :

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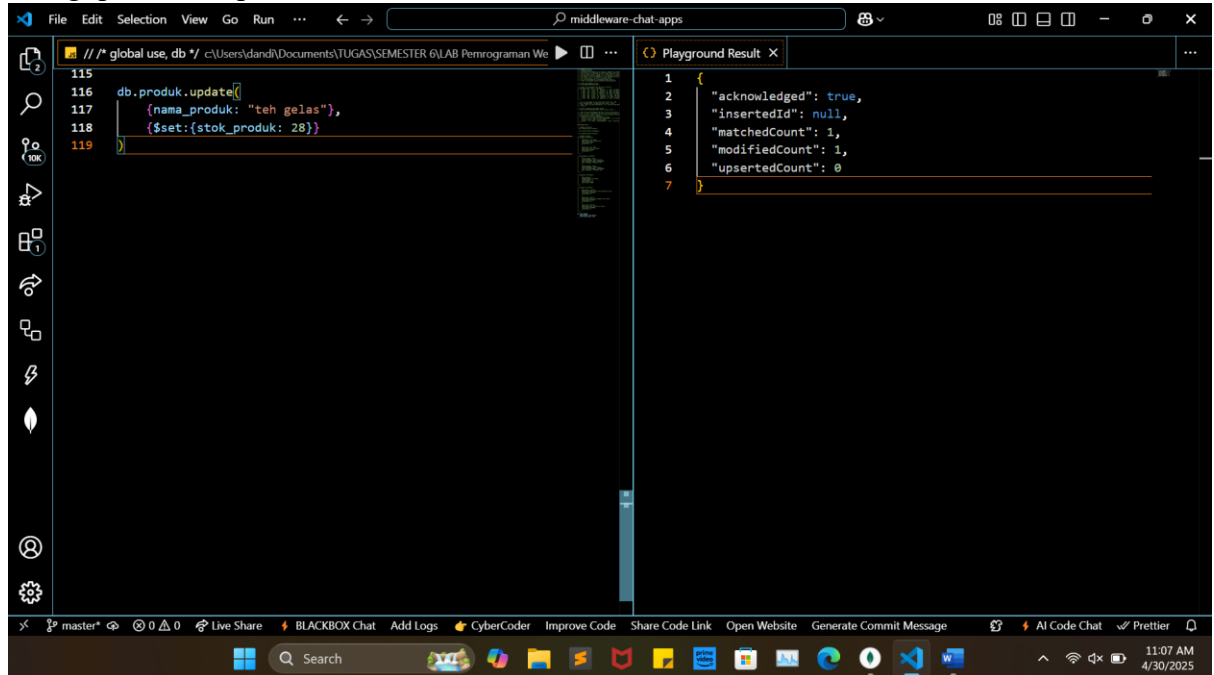
Link Repository Github : https://github.com/DandiHendika/Lab_Basis_Data_Lanjut

QUIZ LAB BASIS DATA LANJUT

1. Desain basis data serta collection dan documentnya



b. Mengupdate stok produk



The screenshot shows a VS Code editor with a file named `global use, db */ c:\Users\dandi\Documents\TUGAS\SEMESTER 6\LAB Pemrograman We`. The code in the editor is as follows:

```
115 // global use, db */ c:\Users\dandi\Documents\TUGAS\SEMESTER 6\LAB Pemrograman We
116 db.produk.update(
117   {nama_produk: "teh gelas"},
118   {$set:{stok_produk: 28}}
119 )
```

The right-hand pane shows the 'Playground Result' of the update operation:

```
1 {
2   "acknowledged": true,
3   "insertedId": null,
4   "matchedCount": 1,
5   "modifiedCount": 1,
6   "upsertedCount": 0
7 }
```

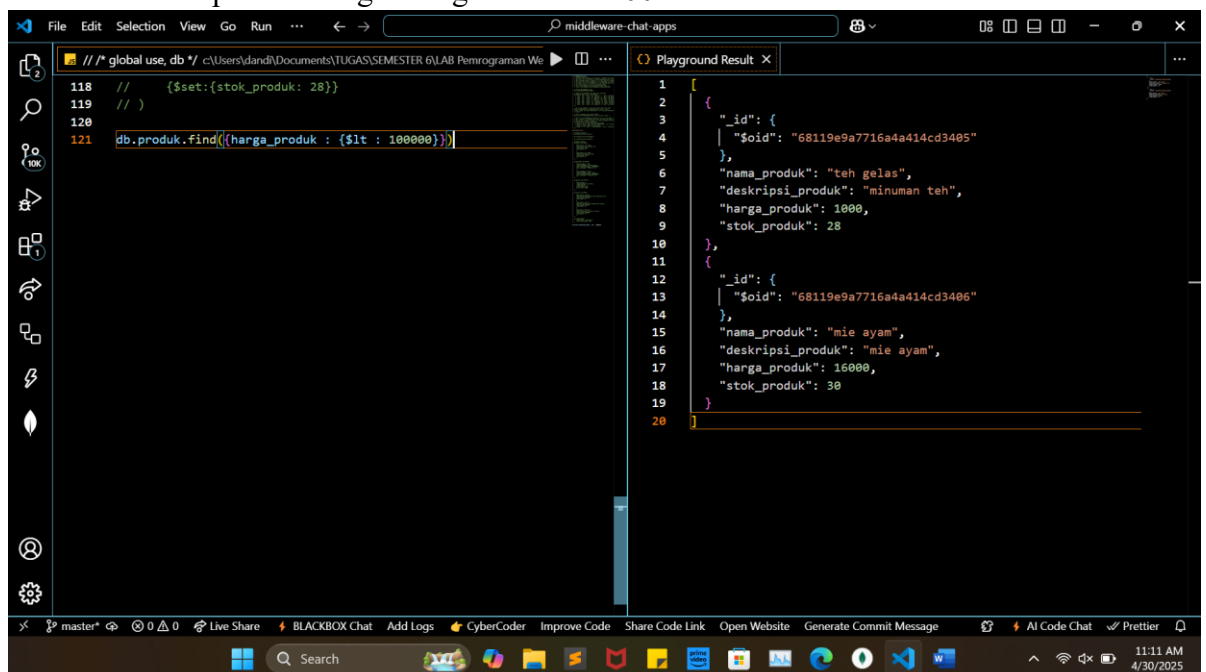
Sebelum:

```
_id: ObjectId('68119e9a7716a4a414cd3405')
nama_produk: "teh gelas"
deskripsi_produk: "minuman teh"
harga_produk: 1000
stok_produk: 30
```

Sesudah:

```
_id: ObjectId('68119e9a7716a4a414cd3405')
nama_produk: "teh gelas"
deskripsi_produk: "minuman teh"
harga_produk: 1000
stok_produk: 28
```

c. Mencari semua produk dengan harga dibawah 100rb



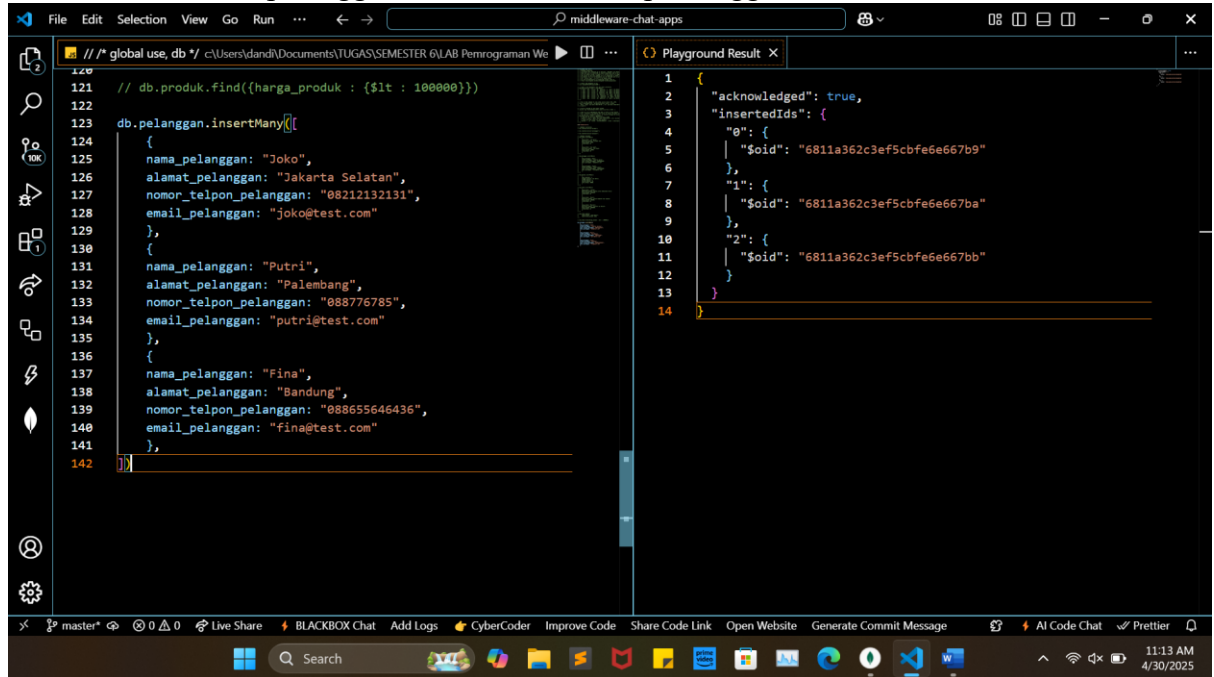
The screenshot shows a VS Code editor with a file named `global use, db */ c:\Users\dandi\Documents\TUGAS\SEMESTER 6\LAB Pemrograman We`. The code in the editor is as follows:

```
118 // {$set:{stok_produk: 28}}
119 // )
120
121 db.produk.find({harga_produk : {$lt : 100000}})
```

The right-hand pane shows the 'Playground Result' of the find operation:

```
1 [
2   {
3     "_id": {
4       "$oid": "68119e9a7716a4a414cd3405"
5     },
6     "nama_produk": "teh gelas",
7     "deskripsi_produk": "minuman teh",
8     "harga_produk": 1000,
9     "stok_produk": 28
10  },
11  {
12    "_id": {
13      "$oid": "68119e9a7716a4a414cd3406"
14    },
15    "nama_produk": "mie ayam",
16    "deskripsi_produk": "mie ayam",
17    "harga_produk": 16000,
18    "stok_produk": 30
19  }
20 ]
```

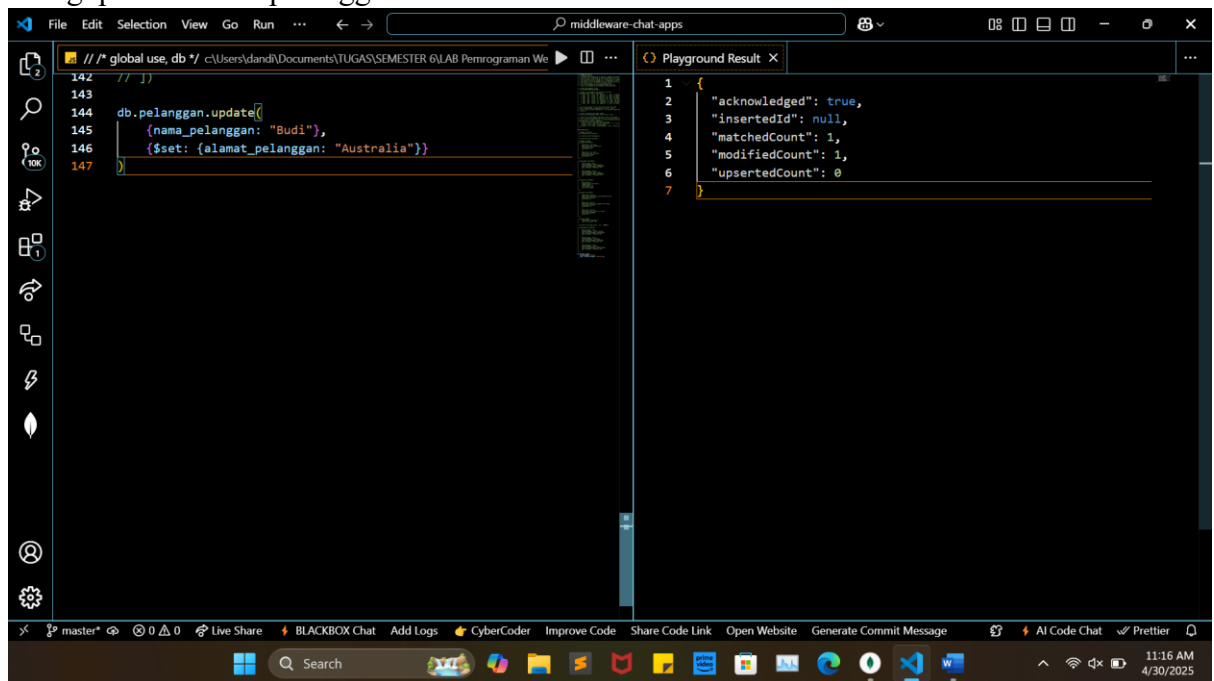
d. Menambahkan data pelanggan baru ke collection pelanggan



```
120 // global use, db */ c:\Users\dandi\Documents\TUGAS\SEMESTER 6\LAB Pemrograman We
121 // db.produk.find({harga_produk : {$lt : 100000}})
122
123 db.pelanggan.insertMany([
124   {
125     nama_pelanggan: "Joko",
126     alamat_pelanggan: "Jakarta Selatan",
127     nomor_telpon_pelanggan: "08212132131",
128     email_pelanggan: "joko@test.com"
129   },
130   {
131     nama_pelanggan: "Putri",
132     alamat_pelanggan: "Palembang",
133     nomor_telpon_pelanggan: "088776785",
134     email_pelanggan: "putri@test.com"
135   },
136   {
137     nama_pelanggan: "Fina",
138     alamat_pelanggan: "Bandung",
139     nomor_telpon_pelanggan: "08865564636",
140     email_pelanggan: "fina@test.com"
141   },
142 ])
```

```
1 {
2   "acknowledged": true,
3   "insertedIds": {
4     "0": {
5       "$oid": "6811a362c3ef5cbfe6e667b9"
6     },
7     "1": {
8       "$oid": "6811a362c3ef5cbfe6e667ba"
9     },
10    "2": {
11      "$oid": "6811a362c3ef5cbfe6e667bb"
12    }
13  }
14 }
```

e. Mengupdate alamat pelanggan



```
142 // }
143
144 db.pelanggan.update(
145   {nama_pelanggan: "Budi"},
146   {$set: {alamat_pelanggan: "Australia"}}
147 )
```

```
1 {
2   "acknowledged": true,
3   "insertedId": null,
4   "matchedCount": 1,
5   "modifiedCount": 1,
6   "upsertedCount": 0
7 }
```

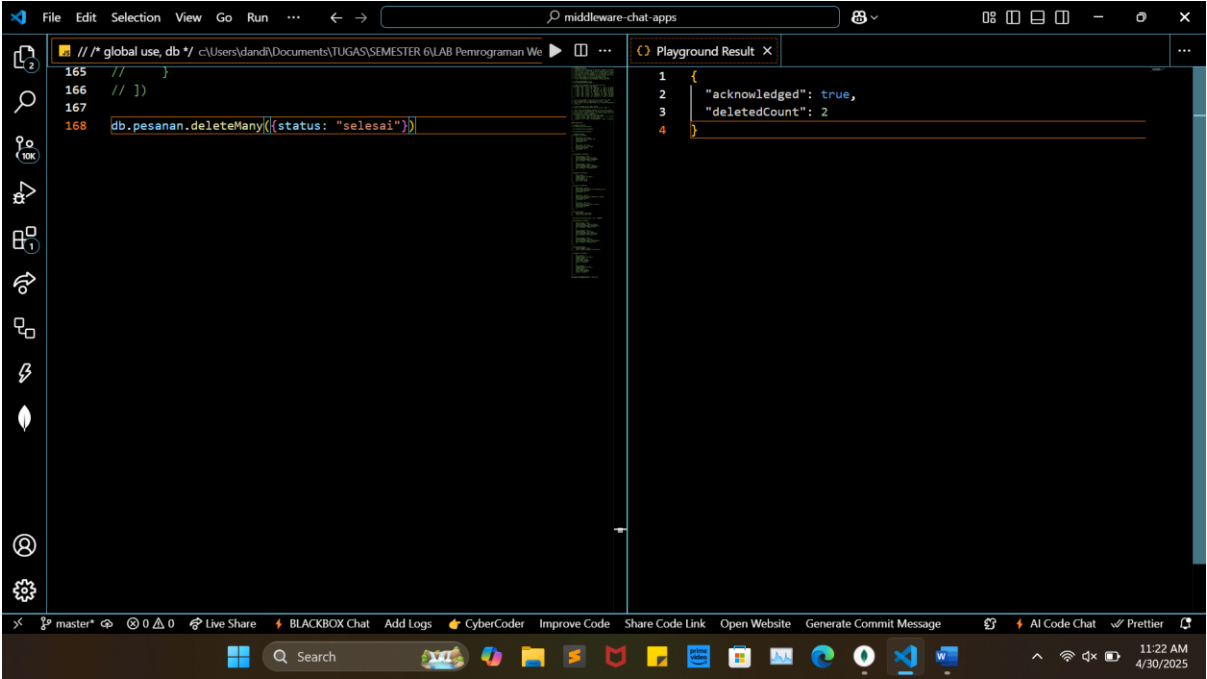
Sebelum:

```
_id: ObjectId('68119d2bbd2e778e6e514d1f')
nama_pelanggan : "Budi"
alamat_pelanggan : "Jakarta barat"
nomor_telpon_pelanggan : "0899999999"
email_pelanggan : "budi@test.com"
```

Sesudah:

```
_id: ObjectId('68119d2bbd2e778e6e514d1f')
nama_pelanggan : "Budi"
alamat_pelanggan : "Australia"
nomor_telpon_pelanggan : "0899999999"
email_pelanggan : "budi@test.com"
```

f. Menghapus pesanan yang sudah selesai dari collection



Sebelum:

pesanan					
	_id ObjectId	nomor_pesanan Int32	tanggal_pesanan Date	produk_dipesan Int32	j
1	ObjectId('68119e6d0196c11...	1	2025-04-30T03:52:13.421+0...	1	1
2	ObjectId('6811a4f2c3b7bf3...	5	2025-04-30T04:20:02.986+0...	5	1
3	ObjectId('6811a4f2c3b7bf3...	6	2025-04-30T04:20:02.986+0...	3	1

Sesudah:

pesanan					
	_id ObjectId	nomor_pesanan Int32	tanggal_pesanan Date	produk_dipesan Int32	j
1	ObjectId('68119e6d0196c11...	1	2025-04-30T03:52:13.421+0...	1	1

3. Perintah MongoDB untuk membuat indeks pada kolom tanggal_pesanan

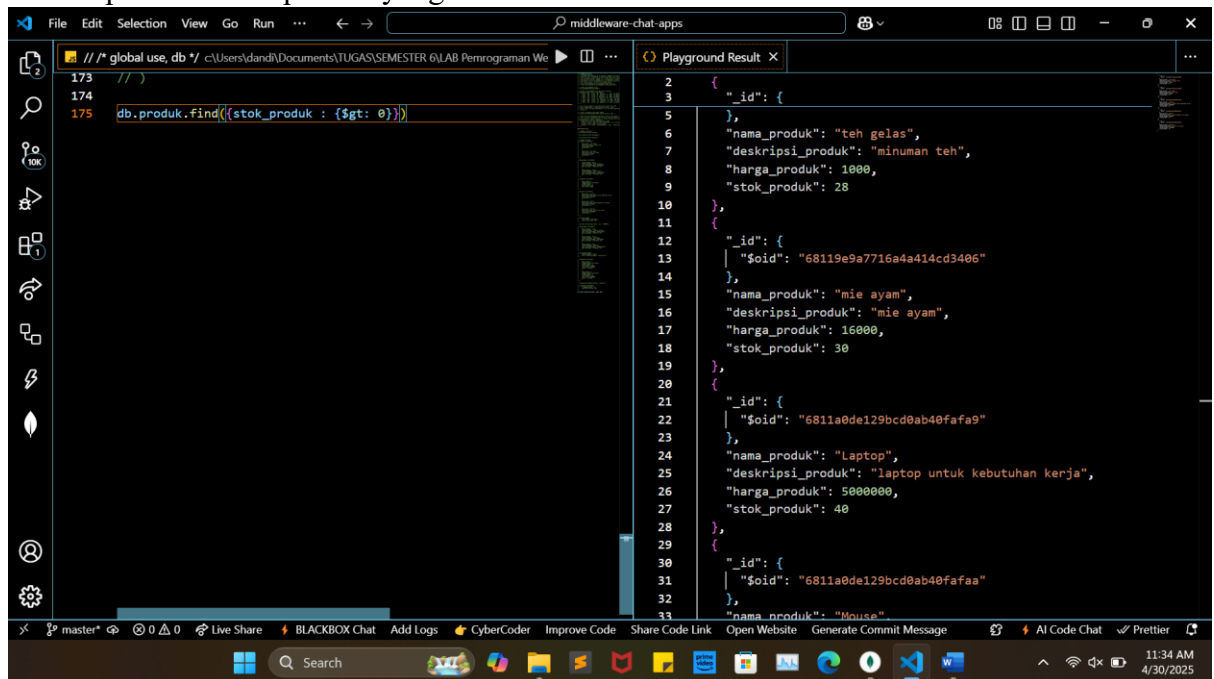
The image shows two windows from a Windows desktop. The top window is Visual Studio Code, displaying a JavaScript file named 'middleware-chat-apps.js'. The code defines a middleware function that creates an index on the 'tanggal_pesanan' field of the 'pesanan' collection in the 'tokoonline' database. The code is as follows:

```
169 // * global use, db */
170 db.pesanan.createIndex({
171   {"tanggal_pesanan":1},
172   {expireAfterSecond: 120}
173 })
```

The bottom window is MongoDB Compass, showing the 'pesanan' collection in the 'tokoonline' database on 'localhost:27017'. The 'Indexes' tab is selected, displaying a table of indexes:

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR	36.9 KB	8 (since Wed Apr 30 2025)	UNIQUE	READY
> tanggal_pesanan_1	REGULAR	20.5 KB	0 (since Wed Apr 30 2025)		READY

4. Output dari operasi berikut:
- Menampilkan semua produk yang tersedia



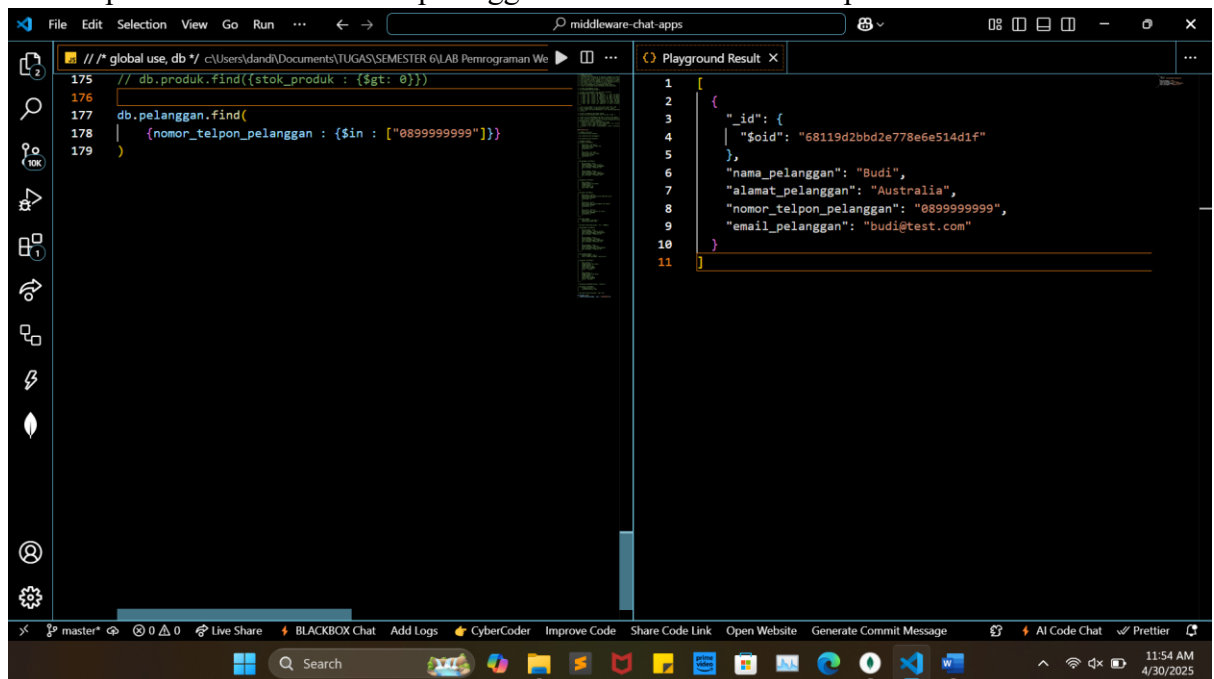
The screenshot shows a VS Code editor with a file named `global use, db */ c:\Users\dandi\Documents\TUGAS\SEMESTER 6\LAB Pemrograman We`. The code in the editor is as follows:

```
173 //  
174  
175 db.produk.find({stok_produk : {$gt: 0}})
```

The `Playground Result` pane on the right displays the output of the query, which is a JSON array of product documents:

```
2 {  
3   "_id": {  
4     "$oid": "68119e9a7716a4a414cd3406"  
5   },  
6   "nama_produk": "teh gelas",  
7   "deskripsi_produk": "minuman teh",  
8   "harga_produk": 1000,  
9   "stok_produk": 28  
10 },  
11 {  
12   "_id": {  
13     "$oid": "6811a0de129bcd0ab40fafa9"  
14   },  
15   "nama_produk": "mie ayam",  
16   "deskripsi_produk": "mie ayam",  
17   "harga_produk": 16000,  
18   "stok_produk": 30  
19 },  
20 {  
21   "_id": {  
22     "$oid": "6811a0de129bcd0ab40fafa9"  
23   },  
24   "nama_produk": "Laptop",  
25   "deskripsi_produk": "laptop untuk kebutuhan kerja",  
26   "harga_produk": 5000000,  
27   "stok_produk": 40  
28 },  
29 {  
30   "_id": {  
31     "$oid": "6811a0de129bcd0ab40fafa9"  
32   },  
33   "nama_produk": "Mouse"
```

- Menampilkan nama dan alamat pelanggan berdasarkan nomor telpon



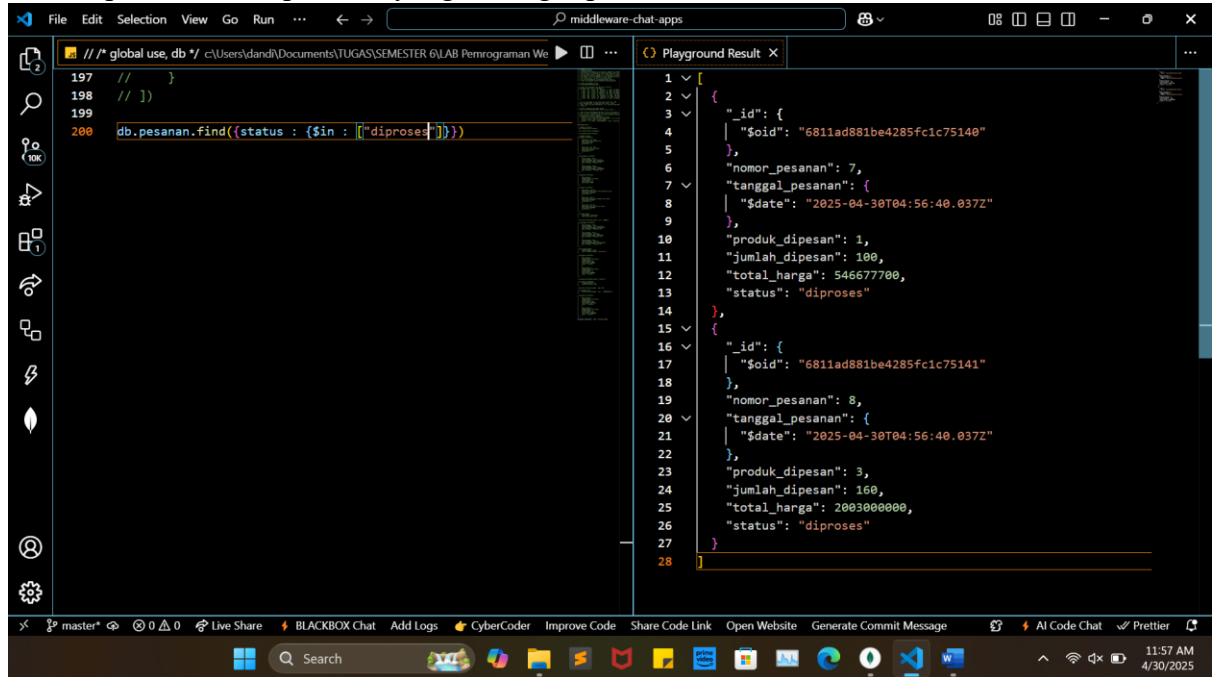
The screenshot shows a VS Code editor with a file named `global use, db */ c:\Users\dandi\Documents\TUGAS\SEMESTER 6\LAB Pemrograman We`. The code in the editor is as follows:

```
175 // db.produk.find({stok_produk : {$gt: 0}})  
176  
177 db.pelanggan.find(  
178   {nomor_telpon_pelanggan : {$in : ["0899999999"]}}  
179 )
```

The `Playground Result` pane on the right displays the output of the query, which is a JSON array of customer documents:

```
1 {  
2   "_id": {  
3     "$oid": "68119d2bbd2e778e6e514d1f"  
4   },  
5   "nama_pelanggan": "Budi",  
6   "alamat_pelanggan": "Australia",  
7   "nomor_telpon_pelanggan": "0899999999",  
8   "email_pelanggan": "budi@test.com"  
9 }  
10  
11 }
```

c. Menampilkan semua pesanan yang sedang diproses



The screenshot shows a VS Code editor with a file named `middleware-chat-apps`. The editor is open to a file with the following code:

```
197 // }  
198 // })  
199  
200 db.pesanan.find({status : {$in : ["diproses"]}})}
```

On the right side, the `Playground Result` tab is open, displaying the results of the MongoDB query. The results are two JSON documents:

```
1 {  
2   "_id": {  
3     "$oid": "6811ad881be4285fc1c75140"  
4   },  
5   "nomor_pesanan": 7,  
6   "tanggal_pesanan": {  
7     "$date": "2025-04-30T04:56:40.037Z"  
8   },  
9   "produk_dipesan": 1,  
10  "jumlah_dipesan": 100,  
11  "total_harga": 546677700,  
12  "status": "diproses"  
13 },  
14  
15 {  
16   "_id": {  
17     "$oid": "6811ad881be4285fc1c75141"  
18   },  
19   "nomor_pesanan": 8,  
20   "tanggal_pesanan": {  
21     "$date": "2025-04-30T04:56:40.037Z"  
22   },  
23   "produk_dipesan": 3,  
24   "jumlah_dipesan": 160,  
25   "total_harga": 2003000000,  
26   "status": "diproses"  
27 },  
28 }
```

The bottom status bar shows the file is named `master` and includes various icons for Live Share, BLACKBOX Chat, Add Logs, CyberCoder, Improve Code, Share Code Link, Open Website, Generate Commit Message, AI Code Chat, and Prettier. The system clock in the bottom right corner indicates the time is 11:57 AM on 4/30/2025.